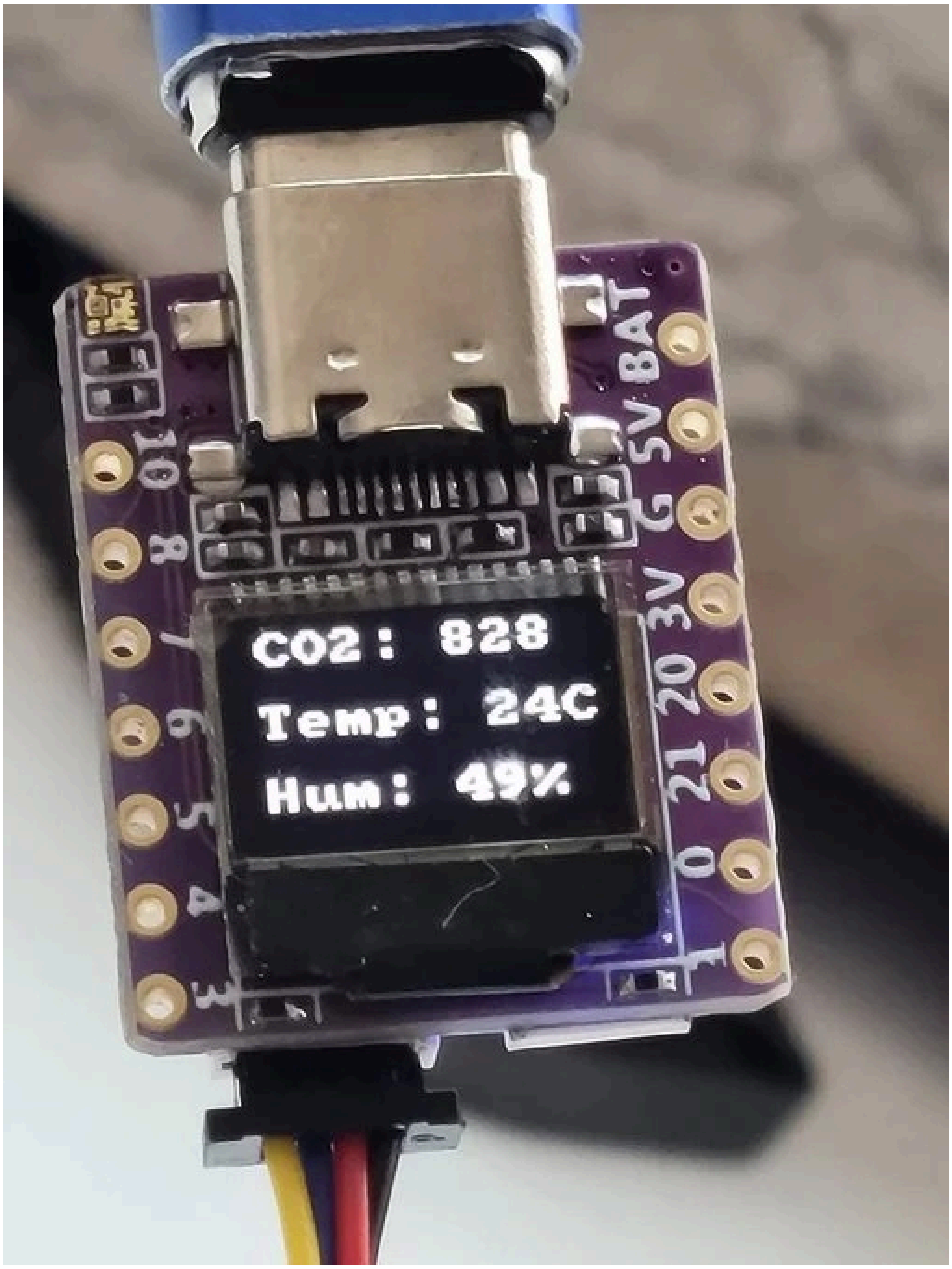
nRF Connect SDK - 3.0.1

Contents

1. ESP32C3 0.42 OLED

Open on GitHub

Board Overview



ESP32C3 0.42 OLED

Name:

`esp32c3_042_oled`

Vendor:

01Space

Architecture:

riscv

SoC:

esp32c3

Overview

ESP32C3 0.42 OLED is a mini development board based on the Espressif ESP32-C3 [1] RISC-V WiFi/Bluetooth dual-mode chip.

For more details see the 01space ESP32C3 0.42 OLED [2] Github repo.

Hardware

This board is based on the ESP32-C3-FH4 with WiFi and BLE support. It features:

- RISC-V SoC @ 160MHz with 4MB flash and 400kB RAM
- WS2812B RGB serial LED
- 0.42-inch OLED over I2C
- Qwiic I2C connector
- One pushbutton
- Onboard ceramic chip antenna
- On-chip USB-UART converter

> **Note**
>
> The RGB led is not supported on this Zephyr board yet.

> **Note**
>
> The ESP32-C3 does not have native USB, it has an on-chip USB-serial converter instead.

Supported Features

The 01space ESP32C3 0.42 OLED board configuration supports the following hardware features:

Interface	Controller	Driver/Component
PMP	on-chip	arch/riscv
INTMTRX	on-chip	intc_esp32c3
PINMUX	on-chip	pinctrl_esp32
USB UART	on-chip	serial_esp32_usb
GPIO	on-chip	gpio_esp32
UART	on-chip	uart_esp32
I2C	on-chip	i2c_esp32
SPI	on-chip	spi_esp32_spim
RADIO	on-chip	Bluetooth
DISPLAY	off-chip	display

Connections and IOs

See the following image:

01space ESP32C3 0.42 OLED Pinout

It also features a 0.42 inch OLED display, driven by a SSD1306-compatible chip. It is connected over I2C: SDA on GPIO5, SCL on GPIO6.

Prerequisites

Espressif HAL requires WiFi and Bluetooth binary blobs. Run the command below to retrieve those files.

```
west blobs fetch hal_espressif
```

> **Note**
>
> It is recommended running the command above after `west update`.

Programming and Debugging

Standalone application

The board can be loaded using a single binary image, without 2nd stage bootloader. It is the default option when building the application without additional configuration.

> **Note**
>
> This mode does not provide any security features nor OTA updates.

Use the following command to build a sample hello_world application:

```
# From the root of the zephyr repository
west build -b esp32c3_042_oled samples/hello_world
```

Sysbuild

Sysbuild (System build) makes it possible to build and flash all necessary images needed to bootstrap the board.

By default, the ESP32 sysbuild configuration creates bootloader (MCUboot) and application images.

To build the sample application using sysbuild, use this command:

```
west build -b esp32c3_042_oled --sysbuild samples/hello_world
```

Flashing

For the `Hello, world!` application, follow the instructions below. Assuming the board is connected to `/dev/ttyACM0` on Linux.

```
# From the root of the zephyr repository
west build -b esp32c3_042_oled samples/hello_world
west flash --esp-device /dev/ttyACM0
```

Since the Zephyr console is by default on the `usb_serial` device, we use the espressif monitor utility to connect to the console.

```
$ west espressif monitor -p /dev/ttyACM0
```

After the board has automatically reset and booted, you should see the following message in the monitor:

```
***** Booting Zephyr OS vx.x.x-xxx-gxxxxxxxxxxxx *****
Hello World! esp32c3_042_oled
```

References

[1]

https://www.espressif.com/en/products/socs/esp32-c3

[2]

https://github.com/01Space/ESP32-C3-0.42LCD